the Vercel AI SDK, provides more specialized tools for building composable conversational chains and agents, and offers practical examples in its documentation. Part of this framework ecosystem also contains tools like LangSmith for building production-grade LLMs and LangGraph for creating agent and multi-agent workflows.

1.4 Choosing the right model

In addition to frontend and backend tooling, we also have to select an LLM model and provider to power our app. For the projects in this book, we'll use several popular models and providers, including

- *OpenAI*—Includes DALL-E 2 for image generation and ChatGPT for language tasks
- *Google AI*—Includes access to its own models, such as Gemini, and integrations with the Vercel AI SDK

There are many considerations when choosing an LLM model and planning how data will flow through the system. Say you're building an app that needs to integrate an LLM model to personalize website content based on user data and preferences. Let's look at the key decisions you'll face. To find the right model for the app, you first need to select the right *type* of model. In this case, you want to select one capable of understanding user preferences and content characteristics. In addition to the model's architecture and approach, the quality and relevance of the model's training data significantly affect its performance. So, you'll want to assess how well the model's training data aligns with your application's specific needs.

1.4.1 Model types

Different types of models excel at different tasks. LLMs are well suited for tasks involving text generation and manipulation, but specific architectures and approaches may be more appropriate for other kinds of tasks:

- *Generative adversarial networks* (GANs) use two competing neural networks to refine image generation so you can create new images based on image inputs. GANs are used to create images on the These Cats Do Not Exist website (https://thesecatsdonotexist.com/) and can also be adapted to generate video.
- *Autoregressive models* predict the next token in a sequence and can generate text one word at a time by predicting the next word in a sentence. This approach can be applied to data other than text and is appropriate for tasks where sequence matters, such as code and music generation, language translation, and time-series prediction. GPT-4 from OpenAI is an autoregressive model.
- *Transformer-based models* analyze relationships within data to excel at tasks like text generation. These models are based on attention mechanisms, allowing them to analyze relationships between different parts of the data. Use transformers when you need global context and large-scale LLM capabilities. Google Bard is an application built on transformer-based LLMs.